

Amirhossein Ghorbani

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EDUCATION

University of British Columbia

Ph.D in Mechanical Engineering, GPA: 90.8/100

– Supervisor: Dr. Sina Kheirkhah

Kelowna, BC, CA

2023–2027

University of Tehran

B.S. in Mechanical Engineering, GPA: 3.85/4.00 (17.93/20)

– GPA of the last two years: 4/4

– Supervisor: Dr. Alireza Jalali

Thesis Score: 4/4 (19.13/20)

Tehran, Iran

2018–2022

RESEARCH INTERESTS

- Combustion
- Microfluidics
- Computational Fluid Dynamics (CFD)
- HVACR
- Turbo-machinery
- Turbulence
- Energy
- Bio-fluids
- Heat & mass transfer
- Aerospace

EXPERIENCE

- **Teaching Assistant (TA) / University of British Columbia** 2023-Present
 - Thermodynamics I (APSC 252)
 - Combustion Processes (ENGR 497)
 - Aircraft Structures (ENGR 449)
 - Fluid mechanics I & II (APSC 253- ENGR 310)
 - Heat transfer applications (ENGR 385)
 - Matter and Energy I & II (APSC 182-183)
 - Energy system design (ENGR 375)
 - Laser-based measurements (ENGR 479)
- **Teaching Assistant (TA) / University of Tehran** 2018-2022
 - Physics 1 | chief TA; quiz & homework designer; part-time teacher
 - Magnetism | quiz homework designer; part-time teacher
 - Fluid mechanics 1 | chief TA; quiz & homework designer; part-time teacher
 - Heat transfer 1 | quiz & homework designer; part-time teacher
 - CFD | quiz & homework designer; part-time teacher
- **Research Assistant (RA) / University of British Columbia** 2023-Present
 - Combustion for Propulsion and Power laboratory | Laminar flame speed of sustainable aviation fuels
- **Research Assistant (RA) / University of Tehran** 2020-2022

SKILLS

- **SOLIDWORKS** (Advanced user)
- **ANSYS Fluent** (Advanced user)
- **MATLAB** (Competed user)
- **COMSOL** (Basic user)
- **EES** (Advanced user)
- **CATIA** (Basic user)
- **MSC Adams** (Competed user)
- **Python** (Basic user)
- **C++** (Competed user)
- **Microsoft Office** (Advanced user)
- **L^AT_EX** (Competed user)

LANGUAGES

- **English** Advanced
 - **IELTS:** 7.5 (L:8.5, R:8.5, W:6.5, S:7)
- **Persian:** Native
- **Arabic:** Basic

HONORS AND AWARDS

- | | |
|---|------|
| • Ranked within the top 1% students amongst more than 160000 participants in Iranian University Entrance Exam. | 2018 |
| • Awarded by the School of Mechanical Engineering, University of Tehran for standing amongst top 10% undergraduate students. | 2021 |
| • Awarded by the University of Tehran for standing amongst top 15 undergraduate students (amongst 132 students). | 2022 |
| • Excellent students' M.Sc. admission offer from the School of Mechanical Engineering, University of Tehran (Exempted from the Iranian Graduate University Entrance Exam plus tuition waiver) | 2022 |

SELECTED COURSES

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| • SOLIDWORKS
<i>University of Tehran (UT) / Scientific Association of Mechanical Engineering (SAME)</i> | 2019 |
| • ANSYS Fluent
<i>Rahnesan Academy</i> | 2020 |
| • Ansys Turbomachinery
<i>Faradars Academy</i> | 2021 |
| • Computational fluid Dynamics (CFD)
<i>Academic professional course / University of Tehran / DR. Alireza Jalali</i> | 2021 |
| • Turbomachinery
<i>Academic professional course / University of Tehran / DR. Mehrdad Raisee</i> | 2021 |
| • Gas Turbine and Jet Propulsion
<i>Academic professional course / University of Tehran / DR. Mehrdad Raisee</i> | 2022 |
| • Design of Refrigeration Systems
<i>Academic professional course / University of Tehran / DR. Behrang Sajadi</i> | 2022 |

REFERENCES

- Dr. Sina Kheirkhah
Associate Professor / Faculty of Engineering / University of British Columbia
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- Dr. Alireza Jalali
Assistant Professor / Faculty of Mechanical Engineering / University of Tehran
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- Dr. Behrang Sajadi
Associate Professor / Faculty of Mechanical Engineering / University of Tehran
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- Dr. Azadeh Jafari
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